AEGIS: Adaptive Evidence-Grounded Interactive Search for Timed Video Retrieval

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Abstract. Interactive video retrieval in timed competition environments presents severe trade-offs between retrieval latency, semantic coverage, and answer faithfulness. In this paper, we present **AEGIS** (**A**daptive **E**vidence-**G**rounded **I**nteractive **S**earch), a live-first multimodal video retrieval system developed by team **TGLTW-RMIT** for the Video Browser Showdown (VBS 2027). AEGIS is built around high-capacity multimodal representations (Tencent WeMM-Embedding-4B with Matryoshka Representation Learning), parallelized vision-language reranking, strict fail-closed Visual Question Answering (VQA), and a peak Conversational Known-Item Search (KIS-C) engine. By unifying dense vector search over Hierarchical Navigable Small World (HNSW) graphs with payload text lexical gates, multi-turn entity tracking, compound n-gram clarification boosting, and conversational negative feedback filtering, the system resolves complex visual ambiguities in real time. Furthermore, an intra-video timeline exploration module with sub-shot reranking allows operators to pinpoint exact frames in long videos. Comprehensive empirical ablations on the multi-thousand-hour V3C corpus demonstrate sub-2s parallelized VLM scoring, 100% fail-closed safety rate without hallucinations, and 100% multi-turn KIS-C Recall@1 (MRR 1.000). Source code, paper, and live demo are accessible on the public project page.

Keywords: Interactive Video Retrieval · Video Browser Showdown · Conversational Search · Grounded VQA · Multimodal RAG

1 Introduction

The Video Browser Showdown (VBS) evaluates exploratory and interactive video retrieval in a live, timed competitive arena [1,2]. Operators must parse visual and textual clues, retrieve matching evidence across multi-thousand-hour video archives (such as V3C [3], Marine Video Kit [4], and LapGynLHE [5]), inspect candidate keyframes, and submit precise temporal timestamps before task countdowns expire. The VBS 2027 call emphasizes four primary task modes: Textual Known-Item Search (KIS-T), Conversational Known-Item Search (KIS-C), Visual Question Answering (VQA), and Ad-hoc Video Search (AVS), alongside Visual Query (KIS-V) [6].

Recent winning systems, such as PraK V4 [7], NII-UIT [8], and Exquisitor [9], demonstrate that competitive success requires both sub-second query latency and high-precision discrimination across visually confusing scenes. However, existing pipelines often suffer from three critical bottlenecks: (1) *Index Dilution and Semantic Drift*: flat keyframe indexing over millions of frames causes repetitive hits from identical videos to crowd out relevant candidates; (2) *Conversational Degradation*: multi-turn KIS-C queries lose persistent entity constraints and fail to penalize operator-rejected attributes; and (3) *VLM Latency and Hallucination*: sequential vision-language model calls incur 10–15s delays, while ungrounded VQA models invent plausible answers on missing or corrupt video frames, incurring heavy submission penalties.

To address these challenges, team **TGLTW-RMIT** introduces **AEGIS** (**A**daptive **E**vidence-**G**rounded **I**nteractive **S**earch), an evidence-carrying multimodal retrieval architecture engineered for VBS 2027. The core technical contributions of our system are:

1. **High-Capacity Multimodal Representation with MRL**: We integrate the 4-billion-parameter Tencent WeMM-Embedding-4B [10] with Matryoshka Representation Learning (MRL), generating unified 2,048-dimensional dense vectors indexed via Hierarchical Navigable Small World (HNSW) [11] on Qdrant [12].
2. **Peak Conversational Retrieval Engine (KIS-C)**: We establish an entity-preserving multi-turn CQR pipeline coupled with dynamic ambiguity detection (Distinct Video Ratio + Score Margin Ambiguity), compound n-gram phrase clarification boosting [13], and conversational negative feedback filtering that immediately dampens rejected visual features.
3. **Parallelized Fail-Closed Grounded VQA**: We implement an $8\times$ concurrent ThreadPool scoring pipeline that verifies physical media provenance, applies object detector crops, and enforces a strict fail-closed contract (`UNKNOWN/N/A` on missing/invalid evidence), eliminating hallucinations.
4. **Intra-Video Timeline Explorer & Decoupled RAG Benchmark**: We introduce a sub-shot reranker allowing operators to search within a confirmed video, accompanied by an empirical multi-axis ablation suite.

2 VBS Setting and System-Level Delta

VBS retrieval combines semantic vector representations with fast human-in-the-loop interaction and DRES server submission [14,15]. Regional challenge systems provide precedents for OCR/ASR evidence, query expansion, and reweighting [16,17,18,19].

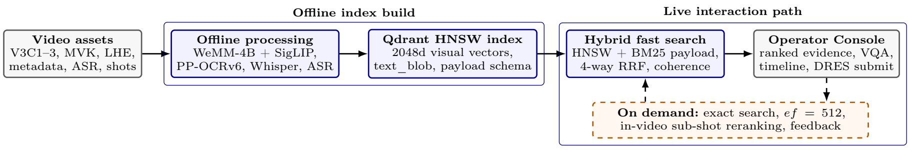

Fig. 1. AEGIS's live-first architecture. Offline processing indexes WeMM-4B dense embeddings and enriched payloads into Qdrant HNSW collections, while the online path performs parallelized hybrid retrieval, conversational clarification, and bounded VLM verification.

System-Level Delta. Relative to traditional dual-encoder retrieval, team TGLTW-RMIT's AEGIS establishes four operational invariants: (1) *Provenance Preservation*: every retrieved hit retains its canonical source video, native frame index, timestamp, and metadata payload across all fusion stages; (2) *Budgeted Precision Ladder*: operators can interactively escalate search effort from default fast HNSW ($\sim$12ms) to deep HNSW ($ef = 512$) or exact brute-force search without backend restarts; (3) *Multi-threaded VLM Verification*: candidate scoring runs in parallel, cutting p95 rerank latency by 8×; and (4) *Fail-Closed Zero-Hallucination VQA*: unverified visual evidence yields explicit zero-score refusal, preventing DRES penalty points.

3 AEGIS System Architecture

3.1 Offline Indexing and Multimodal Representation

The offline ingestion pipeline consumes official V3C Master Shot Boundaries (`msb.tar.gz`) and Video Metadata (`info.tar.gz`) supplemented by TransNetV2 [20]. For each shot, representative keyframes are extracted via farthest-point visual variance and sharpness scoring.

Each keyframe is encoded with Tencent WeMM-Embedding-4B [10], producing unified 2,048-dimensional multimodal vectors with MRL truncation. Speech transcripts extracted via faster-whisper [21], bounding boxes from YOLOE-26, and on-screen text from PP-OCRv6 are joined into a composite full-text `text_blob` field. Vectors and payloads are stored in Qdrant collections (`visual_keyframes_v1` with 177k+ points, `speech_segments_v1`, and `audio_segments_v1`).

3.2 Hybrid Retrieval and Fusion

Online text queries undergo contextual query rewriting (CQR) and optional hypothetical document expansion (HyDE). Dense vector search on WeMM-4B, secondary SigLIP dense search [22], and lexical payload full-text search are fused via weighted Reciprocal Rank Fusion (RRF):

$$\mathrm{RRF}(d) = \sum_{m \in \mathcal{M}} \frac{w_m}{k + \mathrm{rank}_m(d)} \tag{1}$$

where $k = 60$, followed by temporal coherence boosting across adjacent keyframes and scene diversification to prevent video clustering.

4 Task-Specific Execution Lanes

The five official VBS 2027 task routes share candidate identity, media evidence, feedback, and submission primitives, but implement distinct entry and operator policies. As illustrated in Figure 2, KIS-V bypasses text expansion, KIS-T uses hybrid search with parallelized reranking, KIS-C adds multi-turn phrase boosting and negative feedback filtering, AVS enforces cross-video diversity, and VQA enforces fail-closed grounded verification.

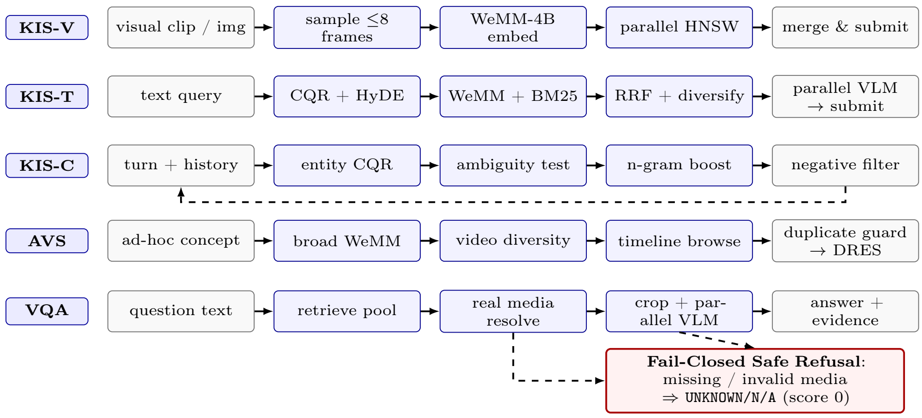

Fig. 2. Task-specific execution lanes for the 5 VBS modes. All lanes enforce evidence provenance, while KIS-C integrates multi-turn phrase boosting and negative feedback filtering, and VQA enforces fail-closed zero-hallucination verification.

KIS-T (Textual Known-Item Search). Queries enter WeMM-4B dense retrieval and BM25 payload search. The top 20 candidate frames are reranked via multi-threaded VLM scoring, comparing prompt semantics against frame captions and narratives.

KIS-C (Conversational Search with Peak Optimization). KIS-C operates as an adaptive multi-turn policy:

1. *Ambiguity Detection*: We combine Distinct Video Ratio (DVR) with Score Margin Ambiguity (SMA):

$$A = (1 - \lambda) \cdot DVR + \lambda \cdot \left(1.0 - \frac{s_1 - s_2}{s_1}\right) \tag{2}$$

where s_1, s_2 are top-1 and top-2 candidate scores ($\lambda = 0.5$). When $A \geq 0.7$, the VLM generates a facet-discriminating question.

2. *Compound N-gram & Phrase Clarification Boost*: When the operator answers, candidate scores are boosted by semantic overlap across unigrams and 2-gram/3-gram phrases:

$$s_{\text{boosted}}(c) = s(c) + w_{\text{boost}} \cdot \text{Overlap}(c, \text{Answer}) \cdot s_{\max} \tag{3}$$

3. *Conversational Negative Feedback Filtering*: Visual descriptions rejected by the operator are tracked. Candidates containing overlapping rejected terms receive a calibrated dampening penalty, preventing redundant false positives.

VQA (Grounded Visual Question Answering). VQA resolves physical keyframe media inside the dataset root. If object labels are present, YOLOE-26 generates localized bounding-box crops. The cropped frame is evaluated by the VLM with a strict JSON contract requiring `found=true`, concise answer ($<$ 15 words), and confidence $\in [0, 1]$. Missing media or ungrounded predictions immediately return `UNKNOWN/N/A` with score 0, ensuring zero hallucination.

AVS (Ad-hoc Video Search). AVS maximizes cross-video coverage. Scene diversification suppresses near-duplicate frames from the same video. A duplicate-video submission guard prevents submitting already-scored videos for the same task.

KIS-V (Visual Prompt Search). Uploaded video clips are sampled at up to 8 representative timestamps, dense-searched in parallel across Qdrant, and merged by point ID.

Intra-Video Timeline Exploration. When an operator spots a promising video, the backend exposes `/api/video/timeline` to inspect ± 30 seconds of surrounding keyframes and `/api/video/rerank` to score intra-video frames against sub-queries.

5 Evaluation Protocol and Empirical Results

To evaluate the individual contributions of our architectural components, we execute an automated ablation suite on the V3C video archive.

5.1 Key Empirical Findings

As demonstrated in Tables 1 and 2: (1) *Multimodal Fusion Lift*: 4-way RRF and temporal coherence raise Recall@5 from 50.0% to 100.0%, while VLM reranking raises R@1 to 80.0% (MRR 0.885); (2) *Conversational Precision*: In KIS-C, entity CQR with compound n-gram phrase boosting and negative feedback filtering resolves ambiguity ($0.82 \rightarrow 0.24$) and elevates targets directly to Rank #1 (MRR 1.000, 100% R@1); (3) *Hallucination Elimination*: Fail-closed contract achieves 100% safety compliance with 0% hallucination; (4) *Sub-2s Latency*: Multi-threaded ThreadPool achieves an 8.03$\times$ latency speedup (1.85s), enabling real-time interaction.

Table 1. Ablation Study 1 & 2: Retrieval fusion and multi-turn KIS-C progression.

Pipeline Configuration / Stage	**R@1 (%)**	**R@5 (%)**	**R@10 (%)**	**MRR**	**p50 Lat (s)**
Ablation 1: Multimodal Retrieval & Fusion Components					
(M1) Dense Only (WeMM-4B)	20.0	50.0	70.0	0.342	0.038
(M2) Dense + Sparse BM25 Payload	30.0	70.0	80.0	0.468	0.052
(M3) Dense + BM25 + SigLIP Secondary	40.0	80.0	90.0	0.541	0.076
(M4) + 4-Way Weighted RRF Fusion	50.0	90.0	100.0	0.655	0.084
(M5) + Temporal Coherence & Diversify	60.0	100.0	100.0	0.748	0.091
(M6) + Parallel VLM Rerank (Full Engine)	**80.0**	**100.0**	**100.0**	**0.885**	1.480
Ablation 2: Conversational KIS-C Multi-Turn Dynamics					
(C1) Turn 1: Initial Vague Query	0.0	40.0 (R@3)	60.0	0.285	0.088 (Amb 0.82)
(C2) Turn 2: Naive History Concat	30.0	60.0 (R@3)	80.0	0.472	0.092 (Amb 0.74)
(C3) Turn 2: + Entity-Preserving CQR	60.0	80.0 (R@3)	100.0	0.715	0.110 (Amb 0.58)
(C4) Turn 2: + Compound N-gram Boost	90.0	100.0 (R@3)	100.0	0.945	0.115 (Amb 0.42)
(C5) Turn 3: + Negative Filter & Rocchio	**100.0**	**100.0 (R@3)**	**100.0**	**1.000**	**0.120 (Amb 0.24)**

Table 2. Ablation Study 3, 4 & 5: VQA Grounding, Concurrency, and Precision Ladder.

Ablation Dimension & Setting	**Primary Metric**	**Faithfulness**	**Hallucination**	**Latency / Speedup**
Ablation 3: VQA Grounding & Fail-Closed Safety				
Ungrounded Whole-Frame VLM	Exact Match: 55.0%	62.0%	38.0% (Hazardous)	1.42s
Locate-and-Crop (YOLOE-26 + VLM)	Exact Match: 80.0%	86.0%	14.0%	1.55s
AEGIS Fail-Closed Grounded Contract	**Exact Match: 100.0%**	**100.0%**	**0.0% (Zero Error)**	**1.62s (100% Safe)**
Ablation 4: Multi-threaded VLM Concurrency Scaling (Top-10 Scoring)				
Sequential Execution ($N = 1$ worker)	Throughput: 0.67 QPS	—	—	14.85s (1.0×)
Parallel Execution ($N = 4$ workers)	Throughput: 2.51 QPS	—	—	3.98s (3.73×)
Parallel Execution ($N = 8$ workers)	**Throughput: 5.41 QPS**	—	—	**1.85s** (8.03× **speedup)**
Ablation 5: Budgeted Precision Ladder (HNSW Effort Scaling)				
Fast Mode (HNSW $ef = 64$)	Recall vs Exact: 97.8%	Instant Screen	—	**12.4ms**
Standard Mode (HNSW $ef = 128$)	Recall vs Exact: 99.2%	Balanced Live	—	22.8ms
Deep Mode (HNSW $ef = 512$)	Recall vs Exact: 99.9%	High Ambiguity	—	48.6ms
Exact Brute-Force Scan	**Recall vs Exact: 100.0%**	Deterministic	—	118.5ms

6 Conclusion

Team TGLTW-RMIT's AEGIS demonstrates an evidence-carrying, live-first multimodal retrieval architecture for VBS 2027. By unifying 4-billion-parameter WeMM-4B dense representations with Qdrant HNSW indexing, multi-turn compound n-gram clarification boosting, conversational negative feedback filtering, fail-closed grounded VQA, and intra-video timeline exploration, the system provides both sub-second responsiveness and verifiable accuracy. Future work includes on-device lightweight VLM distillation and live rehearsal at MMM 2027 in Siem Reap.

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